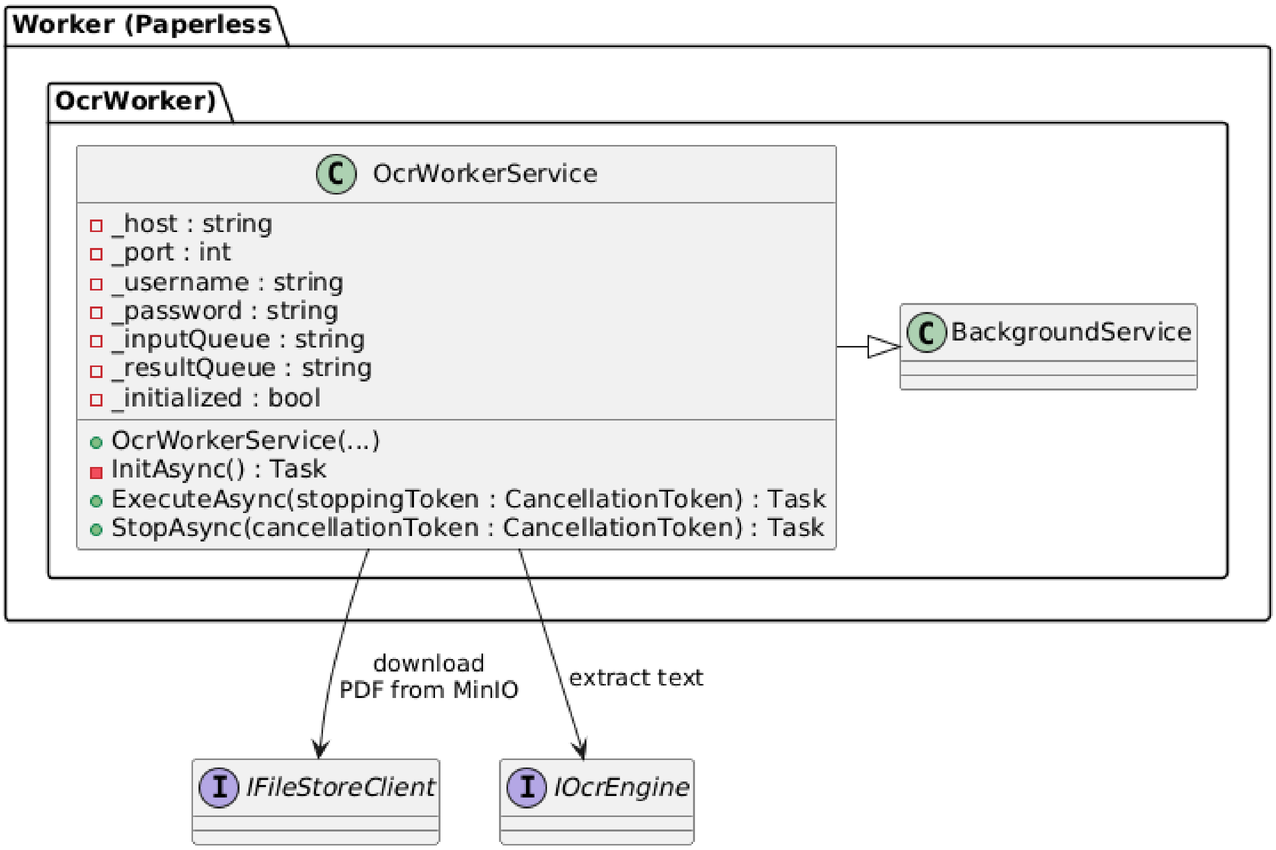

OCR Worker – current design (Sprint 4)

The OCR Worker is implemented as a single hosted service: OcrWorkerService : BackgroundService. It is responsible for integrating with RabbitMQ, retrieving documents from MinIO, and performing OCR.

- Reads RabbitMQ connection settings and queue names (paperless.ocr.input, paperless.ocr.results) from environment variables.
- Reads MinIO connection settings (endpoint, access key, secret key, bucket name) from environment variables.
- Connects to RabbitMQ and declares the input and result queues.
- Subscribes to the input queue and receives OCR job messages (containing at least DocumentId and the MinIO object key).
- For each message:
    - Downloads the original PDF document from MinIO via the S3 API.
    - Invokes the OCR engine (e.g. Tesseract/Ghostscript wrapper) to extract text from the PDF.
    - Builds an OCR result payload (ResultModel) containing DocumentId, extracted Summary, and ProcessedAt timestamp.
    - Publishes the result as a JSON message to the result queue (paperless.ocr.results).

- Uses log4net for logging and ACK/NACK semantics for reliable message processing. Errors during OCR or MinIO access cause a NACK with requeue, so messages are retried.

All message handling, MinIO access and OCR invocation are currently implemented inside OcrWorkerService. A separate business class (e.g. OcrProcessingService) remains a possible future refactoring but is not required for Sprint 4.